



Sara (Fatemeh) KERMANI

PERSONAL DATA AND BIOGRAPHY

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I am a PhD candidate in Computer Science at Sharif University of Technology, specializing in differential privacy. With experience in software engineering (Java/SQL/.NET/C#/C++/Python) and database design. I began programming with C++ during high school and expanded my skills to other languages over time.

INTERESTS

1. *Differential Privacy*
2. *Database*
3. *Data Analysis*
4. *Machine Learning*
5. *Design and Analysis of Algorithms*

WORKING EXPERIENCE

- 2019-Now .NET C# Developer in afarineshdse.ir, an IT corporation and startup studio. Using EF Core Framework, Micro Services, Unit Tests, Integration Tests and Designing SQL Databases for sobherouyesh.com and behkhaan.ir projects. Also, Human Resources Recruitment and Hiring Specialist at afarineshdse.ir.
- SEP - NOV 2024 Database Specialist and Data Analyst at Raydad.com
- 2016-2018 Java and C# Developer in hasintech.com, IT holding
- 2014-2015 Developer in jelodar.com (Not functioning anymore), A Social media and profiling platform for tourism and mountain climbing.

EDUCATION

- 2019-Now PHD in **Computer Science**, **Sharif University of Technology**
Relevant Courses: Optimization for Data Science(17.5), Topics in Optimization(18), Topics in Statistics(17), Statistical Learning(19)
- 2017-2019 M.Sc in Computer Science, Sharif University of Technology
Relevant Courses: Sketching Big Data(16)
- 2012-2017 B.Sc in Computer Science (GPA 15.43/20)
MINOR IN ECONOMICS
Relevant Courses: Fundamental of Programming: (19.2/20), Advanced Programming: (19.8/20), Discrete Structure: (17.6/20), Data Structures: (18.3/20), Theory of Languages and Automata: (16.7/20), Data Base (17/20), Linear Algebra: (14.8/20), Design of Algorithms: (18.5/20), Probability and Applications: (14.2/20), Simulation (16.7/20), Theory of Computer Science (19/20), Image Processing (16/20), Mathematical Logic (18/20), Computer Vision (14.9/20)
- MINOR Fundamental of Economics, Micro Economics, Macro Economics, Mathematical Economics, Econometrics, Political Economics, Money and Banking
- FALL 2012 Entered **Sharif University of Technology**, **Computer Science**, Tehran
- 2012 Graduated with Honor from National Organization for the Development of Exceptional Talents (NODET) GPA (18.60/20).

SKILLS AND LANGUAGES

Programming	Java (spring boot, microservices, hibernate, maven), Docker, Python (numpy, pandas, tensorflow, flask, Differential Privacy libraries), .NET C#, C++, Matlab, R, MySQL Advanced SQL, git, Basic Angular, Basic javascript
Recent Course	Attended to a 30-hours Artificial Intelligence Product Management Bootcamp at Nextera Innovation Center (Tehran University) (sep 2024)

RESEARCH EXPERIENCE AND PUBLICATIONS

2023	Differentially Private All-Pairs Shortest Distances for Low Tree-Width Graphs, https://ieeexplore.ieee.org/document/10323834
2017	Presented a talk entitled K-NEIGHBOURHOOD COLOURING in Colouring, Independence and Domination (CID) Conference University of Zielona Gora, Poland
2014-2017	Graph Coloring Under supervision of Prof. S. Akbari, Department of Mathematical Sciences, Sharif University of Technology (SUT).

TEACHING EXPERIENCE

SPRING 2021	Lecturer, Introduction to Linear Algebra at Mathematics and Computer Science department, Sharif University of Technology
FALL 2020	Lecturer, Discrete Mathematics at Mathematics and Computer Science department, Sharif University of Technology
SPRING 2018	Teaching Assistant, Theory of Languages and Automata, J.B. Ebrahimi
SPRING 2016	Teaching Assistant, Advanced Java Programming , H. Boomari
FALL 2015	Teaching Assistant, Theory of Languages and Automata , S. Khazaei
SPRING 2015	Teaching Assistant, Advanced Java Programming , A. Zarei
2012-2015	Several workshops in NODET high schools in Computer Science, Farzanegan{1,4,5,7,8} , Tehran
2012-NOW	Teaching combinatorics, algorithms and programming in the classes for preparing high school students to participate in Olympiad of informatics, Farzanegan {1,2,4}.

INTERESTS AND LANGUAGES

Interests	Mountain Climbing (Member of technical committee, Sharif University hiking club, 2015), Rock-Climbing, Swimming Reading Books, Music
Languages	English (Fluent), Persian (Native), Azari (Native), Spanish (A2)